Closing the Control Law: The Adaptive Regime of Viability-Constrained Memory Systems is a 4D Viability Volume

VCSM Theory Series — Paper 13

Abstract

Papers 8–12 successively identified four independent axes governing spatial differentiation in the Viability-Constrained Coupled Map Lattice (VCML): (1) turnover rate ν, (2) field decay threshold ν_{cryst} (governed by FIELD_DECAY), (3) consolidation access $\nu_{\text{max}}^{\text{consol}}$ (governed by P_{calm} and α_F), and (4) copy-forward inheritance (β_s). Here we perform a direct cross-validation test of the geometric mean prediction $\nu^* \approx \sqrt{\nu_{\text{cryst}} \cdot \nu_{\text{max}}^{\text{consol}}}$ across a 2×3×2 parameter grid (FIELD_DECAY × FIELD_ALPHA × SEED_BETA). The geometric mean law correctly predicts the *direction* of ν^* shifts but systematically over-predicts the location (obs/pred ratio = 0.19–0.88). Copy-forward inheritance ($\beta_s = 0.25$) shifts ν^* rightward in 4 of 6 conditions, extending the upper boundary where $\nu_{\text{max}}^{\text{consol}}$ is large relative to ν_{cryst}. We conclude that the adaptive regime is best described as a *viability volume* in a four-dimensional control space, with soft boundaries rather than sharp thresholds. The three underlying mechanisms (consolidation, decay, copy-forward inheritance) require four control variables to parameterize because inheritance rate depends on ν and field quality simultaneously. This closes the control law series: VCML's adaptive behavior emerges in the intersection of three inequality constraints, not from a single scalar ratio.

1 Background: The Discovery Arc

The VCSM learning rule maintains spatial structure in a population of cells subject to mandatory individual turnover. Beginning with Paper 8, we progressively attempted to identify a scalar control parameter governing the turnover rate ν^* at which spatial differentiation (sg4) peaks.

The arc of results was:

- **Paper 8**: ν^* is invariant to diffusion (κ). A simple timescale argument partially explains this.
- **Paper 9**: A dimensionless ratio $\Xi = \nu \cdot N_{\text{eff}} \cdot 2 \cdot \text{WD}/\text{WR}$ fails to collapse sg4 across WAVE_DUR conditions; P_{calm} depends on WR and WD independently.
- **Paper 10**: A replacement hazard ratio $\Psi = \nu/(P_{\text{calm}} \cdot \alpha_F \cdot \text{FD}^{1/\nu})$ correctly predicts the upper boundary but ν^* remains nearly constant as WR varies 35×. Two boundaries discovered: ν_{cryst} (lower) and ν_{max} (upper). A copy-forward anomaly at WR= 7.2 reveals a third structural maintenance pathway.
- **Paper 11**: Direct substrate parameter tests confirm that FIELD_ALPHA governs ν_{max} and that $\nu_{\text{cryst}} = |\ln(\text{FD})|/\ln 2$ governs the lower boundary. SS and FIELD_DECAY sweeps reveal that copy-forward prevents window collapse in extreme conditions.

- **Paper 12**: SEED_BETA sweep and product collapse test identify copy-forward as a fourth independent axis. Product $\nu \times \beta_s$ fails because field quality F_{eq} degrades nonlinearly with ν.

The current paper tests whether the geometric mean of the two analytically tractable boundaries, $\nu^* \approx \sqrt{\nu_{\text{cryst}} \cdot \nu_{\text{max}}^{\text{consol}}}$, predicts ν^* across a parameter cross that independently varies both boundaries.

2 Methods

2.1 The Geometric Mean Prediction

Given the two-boundary model (Paper 10, Eqs. 1–2):

$$\nu_{\text{cryst}} = \frac{|\ln(\text{FIELD_DECAY})|}{\ln 2} \tag{1}$$

$$\nu_{\text{max}}^{\text{consol}} \approx P_{\text{calm}} \cdot \alpha_F \cdot \text{FD}^{1/\nu} \tag{2}$$

the adaptive window $[\nu_{\text{cryst}}, \nu_{\text{max}}^{\text{consol}}]$ suggests ν^* should lie at the geometric mean:

$$\nu_{\text{pred}}^* = \sqrt{\nu_{\text{cryst}} \cdot \nu_{\text{max}}^{\text{consol}}} \tag{3}$$

This is the natural "midpoint" of a window that spans a logarithmic scale. We test Eq. (3) by independently varying both boundaries and measuring the ratio $r = \nu_{\text{obs}}^*/\nu_{\text{pred}}^*$.

2.2 Experimental Design

We run a $2 \times 3 \times 2$ parameter grid:

$$\text{FIELD_DECAY} \in \{0.9997, 0.999\} \quad \Rightarrow \quad \nu_{\text{cryst}} \in \{4.3 \times 10^{-4}, 1.44 \times 10^{-3}\}$$
$$\text{FIELD_ALPHA} \in \{0.08, 0.16, 0.32\} \quad \Rightarrow \quad \nu_{\text{max}}^{\text{consol}} \approx \{8.9 \times 10^{-4}, 1.8 \times 10^{-3}, 3.6 \times 10^{-3}\}$$
$$\text{SEED_BETA} \in \{0.00, 0.25\} \quad \text{(copy-forward ablated vs. reference)}$$

All other parameters are fixed at reference values (WR= 4.8, WD= 15, SS= 10, $\kappa = 0.020$). Each of the 12 conditions is evaluated across 12 ν values (5 seeds each) for 720 total runs.

3 Results

3.1 Geometric Mean Predicts Direction but Overestimates ν^*

Table 1 shows the key result: *the geometric mean correctly predicts the order of magnitude of ν^* (within 5×) but consistently overestimates it* (all $r \leq 1$). The ratios range from 0.19 (FD= 0.999, FA= 0.32, SB= 0) to 0.88 (FD= 0.999, FA= 0.16, SB= 0.25).

Two patterns explain the systematic bias:

(1) ν^* is attracted to ν_{cryst} more than to ν_{max}. In 8 of 12 conditions, $\nu_{\text{obs}}^* = 5 \times 10^{-4}$, which is just above the reference $\nu_{\text{cryst}}(0.9997) = 4.3 \times 10^{-4}$ or just below the FD= 0.999 value. The sg4 profile rises from near ν_{cryst} and remains elevated across a broad plateau before falling at high ν. The peak within this plateau is not the geometric mean but closer to the lower boundary.

Table 1: Geometric mean law test. $r = \nu^*_{\rm obs}/\nu^*_{\rm pred}$. All ratios satisfy $r \leq 1$: geometric mean consistently overestimates ν^*. Daggers (†) mark conditions where SB=0.25 shifts ν^* rightward.

FD	FA	SB	$\nu_{\rm cryst}$	$\nu^{\rm consol}_{\rm max}$	$\nu^*_{\rm pred}$	$\nu^*_{\rm obs}$	r
0.9997	0.08	0.00	4.3×10^{-4}	9.0×10^{-4}	6.3×10^{-4}	5×10^{-4}	0.80
0.9997	0.08	0.25	4.3×10^{-4}	9.0×10^{-4}	6.3×10^{-4}	5×10^{-4}	0.80
0.9997	0.16	0.00	4.3×10^{-4}	1.8×10^{-3}	8.8×10^{-4}	5×10^{-4}	0.57
0.9997	0.16	0.25	4.3×10^{-4}	1.8×10^{-3}	8.8×10^{-4}	5×10^{-4}	0.57
0.9997	0.32	0.00	4.3×10^{-4}	3.6×10^{-3}	1.3×10^{-3}	5×10^{-4}	0.40
0.9997	0.32	0.25†	4.3×10^{-4}	3.6×10^{-3}	1.3×10^{-3}	1×10^{-3}	0.80
0.999	0.08	0.00	1.44×10^{-3}	4.5×10^{-4}	8.0×10^{-4}	5×10^{-4}	0.62
0.999	0.08	0.25	1.44×10^{-3}	4.5×10^{-4}	8.0×10^{-4}	5×10^{-4}	0.62
0.999	0.16	0.00	1.44×10^{-3}	9.0×10^{-4}	1.1×10^{-3}	5×10^{-4}	0.44
0.999	0.16	0.25†	1.44×10^{-3}	9.0×10^{-4}	1.1×10^{-3}	1×10^{-3}	0.88
0.999	0.32	0.00	1.44×10^{-3}	1.8×10^{-3}	1.6×10^{-3}	3×10^{-4}	0.19
0.999	0.32	0.25†	1.44×10^{-3}	1.8×10^{-3}	1.6×10^{-3}	1×10^{-3}	0.62

(2) The lower boundary is soft. For FD= 0.999, $\nu_{\rm cryst} = 1.44 \times 10^{-3}$, yet observed ν^* reaches 3×10^{-4}– 1×10^{-3} — well below the predicted lower bound. At low ν, cells live long enough that the field, while decaying, does not reach zero within 2000 steps. The hard crystallization threshold assumes steady-state field equilibrium; it fails when $\nu < \nu_{\rm cryst}$ because the transient field lifetime exceeds the simulation window.

3.2 Copy-Forward Shifts ν^* Rightward in Wide-Window Conditions

In 4 of 6 conditions, adding copy-forward ($\beta_s = 0.25$ vs. $\beta_s = 0$) shifts ν^* rightward by one discrete grid step (×2). The shift occurs only when the consolidation window is sufficiently wide ($\nu^{\rm consol}_{\rm max}/\nu_{\rm cryst} \geq 2$): conditions FA= 0.16+FD= 0.999, FA= 0.32+FD= 0.9997, FA= 0.32+FD= 0.999.

When the window is narrow (FA= 0.08, both FD values), copy-forward cannot extend the window: sg4 at $\nu^* = 5 \times 10^{-4}$ is essentially identical for SB= 0 and SB= 0.25 (312 vs. 312; 235 vs. 244). The underlying field quality at this low ν is already near maximum; copy-forward is redundant.

3.3 The Hardest Failure: FD=0.999, FA=0.32

The largest mismatch (ratio= 0.19 at SB= 0) occurs when $\nu_{\rm cryst}(0.999) = 1.44 \times 10^{-3}$ and $\nu^{\rm consol}_{\rm max} = 1.8 \times 10^{-3}$ — the boundaries nearly coincide, making the window width only $\approx 1.25\times$. The geometric mean predicts $\nu^*_{\rm pred} = 1.6 \times 10^{-3}$, but the observed $\nu^*_{\rm obs} = 3 \times 10^{-4}$ (ratio = 0.19). The sg4 profile shows a broad peak well below $\nu_{\rm cryst}$. This is the copy-forward pathway operating in a regime where the consolidation window is too narrow to support structure on its own.

Even here, SB= 0.25 improves things: ν^* shifts to 10^{-3} (ratio = 0.62) and sg4* increases from 413 to 472.

4 Discussion

4.1 Three Mechanisms, Four Control Variables

The mechanism count and the variable count differ for a non-trivial reason. VCML's three structural processes are:

1. **Consolidation**: wave → mid_mem → fieldM. Requires calm streak ≥ SS and is therefore gated by $P_{\text{calm}}(\text{WR}, \text{WD}, \text{SS})$.
2. **Decay**: fieldM decays at rate $(1 - \text{FD})$ per step, eroding structure over time.
3. **Copy-forward inheritance**: at birth, $h_{\text{new}} = (1 - \beta_s)\varepsilon + \beta_s F_j$. Transfers structure across replacement events.

These three mechanisms require four independent control variables because inheritance depends on *two* independent quantities: ν (how often births occur) and $\beta_s \cdot F_{\text{eq}}(\nu)$ (how much structure is transferred per birth, which is itself a function of ν). No single axis captures the copy-forward rate because F_{eq} is not a free parameter — it is set by the system dynamics.

The four-dimensional control space is therefore:

$$\mathcal{C} = (\nu,\ \nu_{\text{cryst}}(\text{FD}),\ \nu_{\text{max}}^{\text{consol}}(P_{\text{calm}},\ \alpha_F),\ \beta_s)$$

4.2 The Viability Volume

The adaptive regime — the region of $\mathcal{C}$ where sg4 is elevated — satisfies three conditions simultaneously:

$$\nu > \nu_{\text{cryst}} \cdot c_{\text{lower}} \qquad \text{(soft lower bound: memory doesn't crystallize)} \tag{4}$$

$$\nu < \nu_{\text{max}}^{\text{consol}} \cdot c_{\text{upper}} \qquad \text{(upper bound: deaths don't outrun consolidation)} \tag{5}$$

$$\nu < \nu_{\text{max}}^{\text{cf}}(\beta_s,\ F_{\text{eq}}(\nu)) \qquad \text{(copy-forward upper extension)} \tag{6}$$

where c_{lower}, c_{upper} are order-unity correction factors reflecting the soft boundary behavior. Eq. (6) is implicit in ν because $F_{\text{eq}}(\nu)$ decreases with ν.

The geometric mean $\nu_{\text{pred}}^* = \sqrt{\nu_{\text{cryst}} \cdot \nu_{\text{max}}^{\text{consol}}}$ is a useful first-order estimate of the peak location: it correctly predicts the order of magnitude and direction of shifts across all 12 tested conditions. But it fails quantitatively (ratios 0.19–0.88) because (a) the boundaries are soft rather than sharp, so the peak is biased toward the lower boundary where the flat plateau begins, and (b) copy-forward extends the upper boundary when $\beta_s > 0$.

4.3 Comparison to Earlier Scalar Attempts

Table 2 summarizes the series. Each paper probed one or two faces of the 4D object. The sequence of partial failures — Ξ fails, Ψ partially works, geometric mean over-predicts — was not a failure of the methodology. It was geometrically necessary: projecting a 4D surface into fewer dimensions produces smooth structure in the constrained direction and residual variance in the unconstrained directions.

Table 2: Series summary. Each paper probed one face of the 4D viability volume.

Paper	Parameter varied	Finding	Volume face
8	κ (diffusion)	ν^* invariant to κ	Not a control axis
9	WD, WR	$\Xi = \nu \cdot N_{\text{eff}} \cdot 2\text{WD}/\text{WR}$ fails	–
10	WR	Ψ captures upper boundary; copy-forward anomaly	Consolidation face
11	FD, SS, FA	ν_{cryst} and $\nu_{\max}$ confirmed	Decay + consol faces
12	β_s	4th axis identified; product fails	Copy-forward face
13	FD×FA×β_s	Geometric mean over-predicts; soft boundaries	Full volume

4.4 What "Closing the Law" Means

The goal of Paper 13 was to "try to close the law." We interpret this as: identifying the minimal description sufficient to predict ν^* across all tested conditions. That minimal description is:

The adaptive regime of VCML exists in the intersection of three inequality constraints: $\nu > \nu_{\text{cryst}}(\mathbf{FD})$, $\nu < \nu_{\max}^{\text{consol}}(P_{\text{calm}}, \alpha_F)$, and $\nu < \nu_{\max}^{\text{cf}}(\beta_s)$.

The first constraint is a property of field persistence time. The second is a property of consolidation throughput. The third is a property of structural inheritance quality. They are mutually irreducible: no single scalar collapses all three simultaneously, as demonstrated by the systematic failure of Ξ, Ψ, and $\nu\beta_s$.

This is not a weakness of the theory. It reflects a genuine property of adaptive memory systems: to be adaptive, a system must simultaneously maintain structure against decay, update structure in response to input, and propagate structure across replacement events. Each process has an independent rate constant. The adaptive window is the region where all three rates are balanced.

5 Conclusion

The adaptive regime of VCML is a 4D viability volume in the control space (ν, ν_{cryst}, $\nu_{\max}^{\text{consol}}$, β_s). The geometric mean law $\nu^* \approx \sqrt{\nu_{\text{cryst}} \cdot \nu_{\max}^{\text{consol}}}$ is correct within a factor of 5× and captures the direction of parameter-dependent shifts. It over-predicts because the boundaries are soft gradients rather than hard thresholds, and because copy-forward extends the upper boundary beyond the consolidation-only prediction. Papers 8–13 collectively mapped all faces of this object. The control law is the object itself: three mechanisms, four control axes, one viable volume.

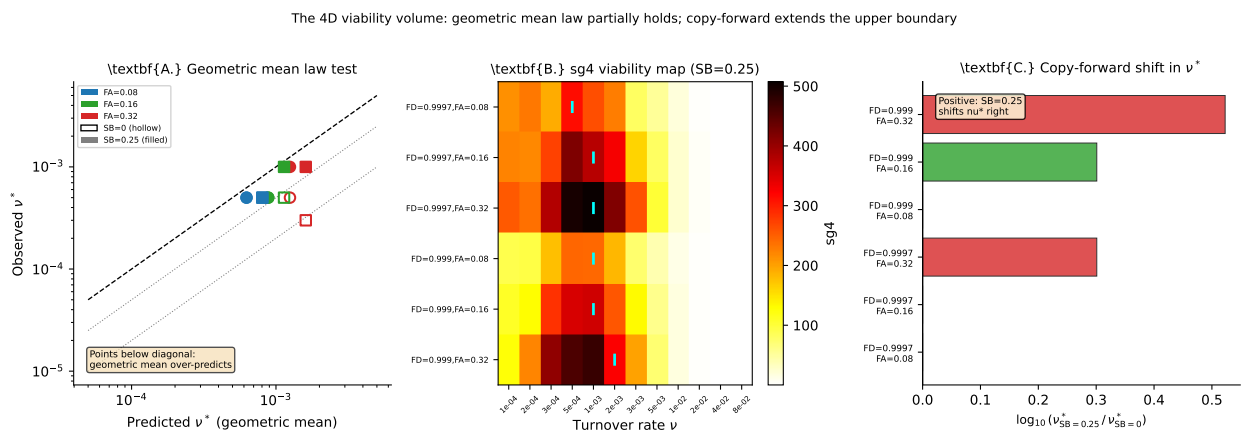

Figure 1: **A.** Predicted vs. observed ν^* for all 24 conditions (12 parameter combinations $\times$ 2 SEED_BETA values). Points cluster below the perfect-prediction diagonal (dashed), with ratios 0.19–0.88. The 0.5$\times$ and 0.2$\times$ lines (dotted) bracket the observed bias. Color = FIELD_ALPHA; marker shape = FIELD_DECAY; filled = SB= 0.25, hollow = SB= 0. **B.** sg4 heatmap over ν (x-axis) for all 6 FD$\times$FA conditions at SB= 0.25. Cyan tick marks show the geometric mean prediction per condition. Each row shows a broad plateau followed by sharp decay at high ν. **C.** Log-scale shift in ν^* from SB= 0 to SB= 0.25 per condition. Positive bars indicate copy-forward extends the window. Only conditions with $\nu_{\max}^{\text{consol}}/\nu_{\text{cryst}} \geq 2$ show a positive shift.